CYBER SECURITY • INFOGRAPHIC PROJECT

SOCIAL ENGINEERING ATTACKS & MAJOR SECURITY BREACHES

Real-world case studies • Attack techniques • Prevention strategies

SUBJECT	CNS (Computer / Network Security)
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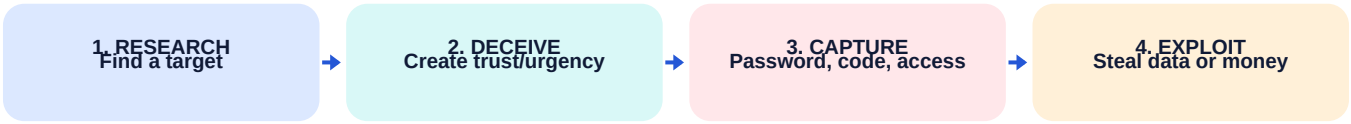
“People are often the first target — awareness is the first defence.”

01 / UNDERSTAND THE ATTACK

Social engineering is the psychological manipulation of people into revealing confidential information, granting access, or taking unsafe actions. Instead of attacking only technology, criminals exploit trust, urgency, fear, curiosity, or helpfulness.

THE COMMON ATTACK FLOW



THREE HIGH-FREQUENCY TECHNIQUES

■ PHISHING

A fake email, SMS, website, or chat pretends to be a trusted service and asks the victim to click a link or enter credentials.

■ PRETEXTING

The attacker invents a believable story or identity — for example, pretending to be IT support, a bank employee, or a manager.

■ BAITING

A tempting offer or item — such as a ‘free’ download, prize, or unknown USB drive — encourages the victim to open a risky file or connect a device.

■ VISHING / IMPERSONATION

A phone call or message impersonates a trusted person and pressures someone to share a one-time code, reset access, or bypass a process.

Why it works: attackers often create urgency (“act now”), authority (“I’m from IT”), fear (“your account will close”), or curiosity (“open this file”). Pause and verify independently.

02 / REAL-WORLD CASE STUDIES

CASE STUDY A — TWITTER, JULY 2020

WHAT HAPPENED

Attackers called employees while pretending to be Twitter IT help-desk staff, claimed there was a VPN problem, and directed some employees to a convincing fake login page. Stolen credentials and authentication approvals helped attackers reach internal tools. High-profile accounts were then taken over and used to promote a Bitcoin scam.

WHY IT SUCCEEDED

The attackers used believable workplace context, employee research, and impersonation. Weaknesses in internal access controls increased the impact after the initial compromise.

HOW IT COULD HAVE BEEN PREVENTED

Verify help-desk identity through a separate trusted channel; use phishing-resistant MFA; train staff to report suspicious calls; restrict privileged tools; and alert on unusual account or admin activity.

CASE STUDY B — MGM RESORTS, SEPTEMBER 2023

WHAT HAPPENED

Public reporting described a social-engineering attack in which criminals reportedly used information about an employee to impersonate them when contacting IT support. The incident disrupted MGM systems and services, illustrating how a help-desk process can become a route into a large organization.

LESSON & PREVENTION

Do not reset passwords or MFA based only on a caller’s confidence or personal details. Require strong identity verification, dual approval for sensitive resets, least-privilege access, rehearsed incident response, and reliable offline recovery procedures.

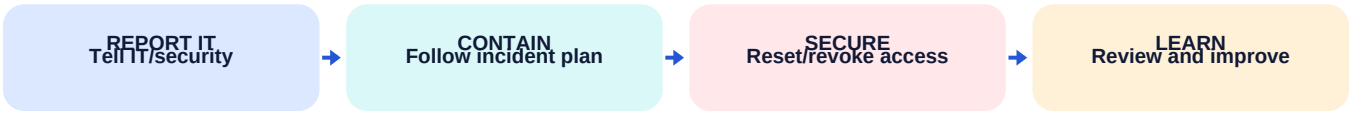
KEY LESSON: A small human-facing weakness can become a large organization-wide incident when privileged access is not tightly controlled.

03 / PREVENTION PLAYBOOK

A practical checklist for students, employees, and organizations

01	**STOP & CHECK** Be suspicious of urgency, threats, unexpected prizes, and unusual requests.
02	**VERIFY THE SENDER** Check the full email address/domain. Contact the person using a known number or official app.
03	**PROTECT CREDENTIALS** Never share passwords, OTPs, recovery codes, or approve an MFA prompt you did not initiate.
04	**CHECK LINKS & FILES** Preview URLs, avoid unknown attachments, and never plug in found USB drives.
05	**USE STRONG SECURITY** Use a password manager, unique passwords, phishing-resistant MFA where available, and updated software.
06	**REPORT QUICKLY** Report suspicious messages immediately. If you clicked, disconnect if instructed by IT, change credentials from a clean device, and contact the service provider.

IF YOU SUSPECT A BREACH:



CONCLUSION

Social engineering succeeds when a request is trusted before it is verified. Combining awareness, identity verification, least privilege, secure authentication, and fast reporting reduces the chance that one mistake becomes a major breach.

REFERENCES / FURTHER READING

• New York State Department of Financial Services, *Twitter Investigation Report*: https://www.dfs.ny.gov/reports-and-publications/other-reports/Twitter_Report

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• Microsoft Security Blog, *Midnight Blizzard conducts targeted social engineering over Microsoft Teams* (2 Aug 2023): https://www.microsoft.com/en-us/security/blog/2023/08/02/midnight-blizzard-conducts-targeted-social-engineering-over-microsoft-teams/

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